

NAME: _____

PERIOD: _____

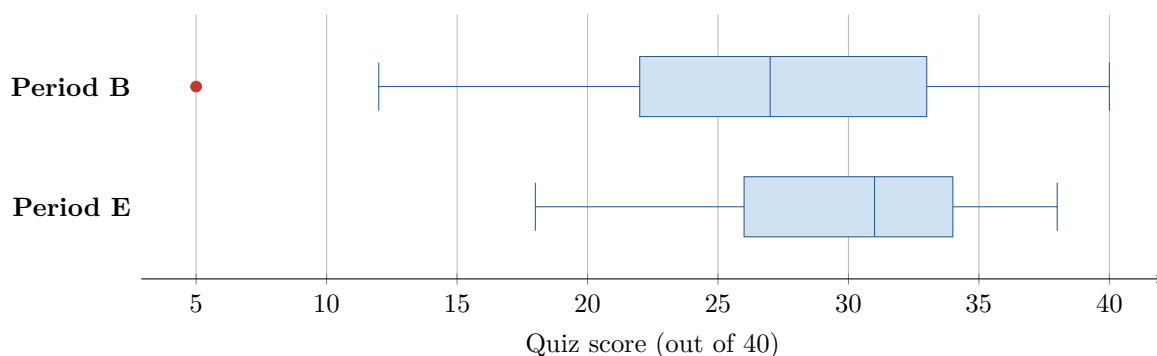
DATE: _____

Which Class Did Better?

AP Statistics · Unit 1 · Topic 1.9 · B: 2026-09-24 · E: 2026-09-30

[STAR] TODAY'S PROBLEM

Two AP Statistics classes took the same 40-point quiz. The side-by-side boxplots show the scores. Two students read them differently. **Sam**: “Period B did better — its highest score is 40 and Period E’s is only 38.” **Tia**: “Period E did better — its median is higher and its scores are more consistent.”



FIRST TAKE — before the video (5 min)

Which class did better on the quiz? One sentence, and name ONE feature of the plot you used.

[BOOK] COMPARING DISTRIBUTIONS (keep on the page)

Compare, don't list: every sentence names BOTH groups (“Period E’s median is higher than Period B’s”).

Cover **center** (median / mean) · **variability** ($IQR = Q3 - Q1$, range) · **shape** (where the median sits in the box, whisker lengths) · **unusual features** (outlier dots) — in context, with units.

A maximum or minimum is ONE student. A median is the typical student.

Finish after the video:

DOK 1 (a) Read off the median and the maximum score for each class.

DOK 2 (b) Compare the two distributions of quiz scores in context. Use comparative language (higher / lower, more / less variable) and address center, variability, shape, and any unusual features.

DOK 3 (c) Whose claim is better supported, Sam's or Tia's? Cite two specific features of the boxplots that settle it. Then: if you could report only ONE statistic per class to a parent asking 'how did the class do?', which statistic would you choose, and why that one rather than the maximum?

Frame: _____'s claim is better supported because the _____ of Period E is _____ than Period B's, and Period E's scores are _____.

Frame: I would report the _____ because the maximum describes _____.

EXIT — turn this in

One thing the video changed about my first take: _____

Optional #1 — not collected

DOK 2 Two bus routes were timed for 20 mornings each. Route 1: mean 24.0 min, SD 6.5 min, median 22 min. Route 2: mean 24.5 min, SD 1.8 min, median 24 min. Write two sentences comparing the routes for a student who must not be late, using comparative language and the statistics given.